

Exploratory Data Analysis (EDA) Report

Objective

To analyze the Titanic dataset using statistical summaries and visualizations to identify patterns, trends, and relationships.

Introduction

Exploratory Data Analysis (EDA) helps understand the dataset before modeling.

Dataset Preview

df.head()

Choose file: titanic.csv

titanic.csv(text/csv) - 61194 bytes, last modified: 02/07/2026 - 100% done

Saving titanic.csv to titanic.csv

First 5 Rows

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cummings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Displays the first five rows of the dataset.

Dataset Information

df.info()

Dataset Information

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 891 entries, 0 to 890

Data columns (total 12 columns):

#	Column	Non-Null	Count	Dtype
---	-----	-----	-----	-----
0	PassengerId	891	non-null	int64
1	Survived	891	non-null	int64
2	Pclass	891	non-null	int64
3	Name	891	non-null	object
4	Sex	891	non-null	object
5	Age	714	non-null	float64
6	SibSp	891	non-null	int64
7	Parch	891	non-null	int64
8	Ticket	891	non-null	object
9	Fare	891	non-null	float64
10	Cabin	204	non-null	object
11	Embarked	889	non-null	object

dtypes: float64(2), int64(5), object(5)

memory usage: 83.7+ KB

Shows columns, data types, and missing values.

Statistical Summary

df.describe()

Statistical Summary							
	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

Summarizes numerical features.

Missing Values

Before cleaning.

Missing Values	
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2
dtype:	int64

Identifies missing values.

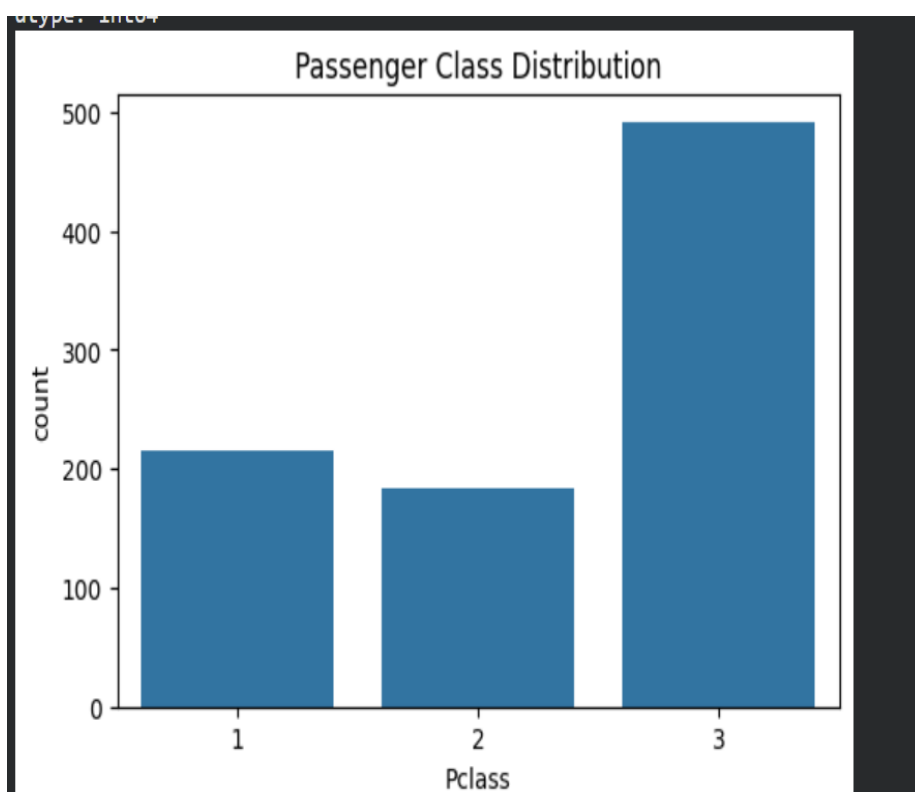
Data Cleaning

After cleaning.

```
Missing Values After Cleaning
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64
```

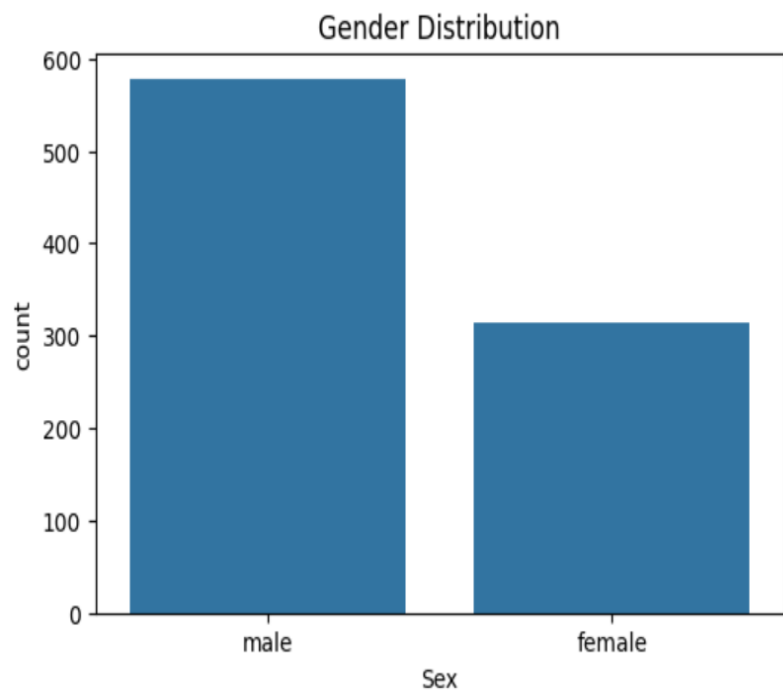
Missing values handled and Cabin removed.

Passenger Class Distribution



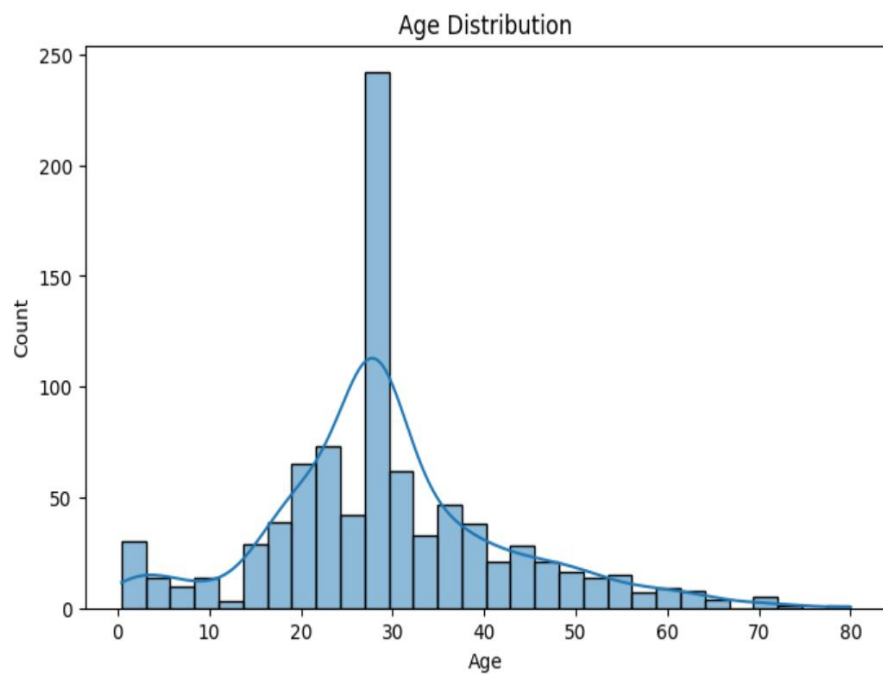
Shows passenger counts by class.

Gender Distribution



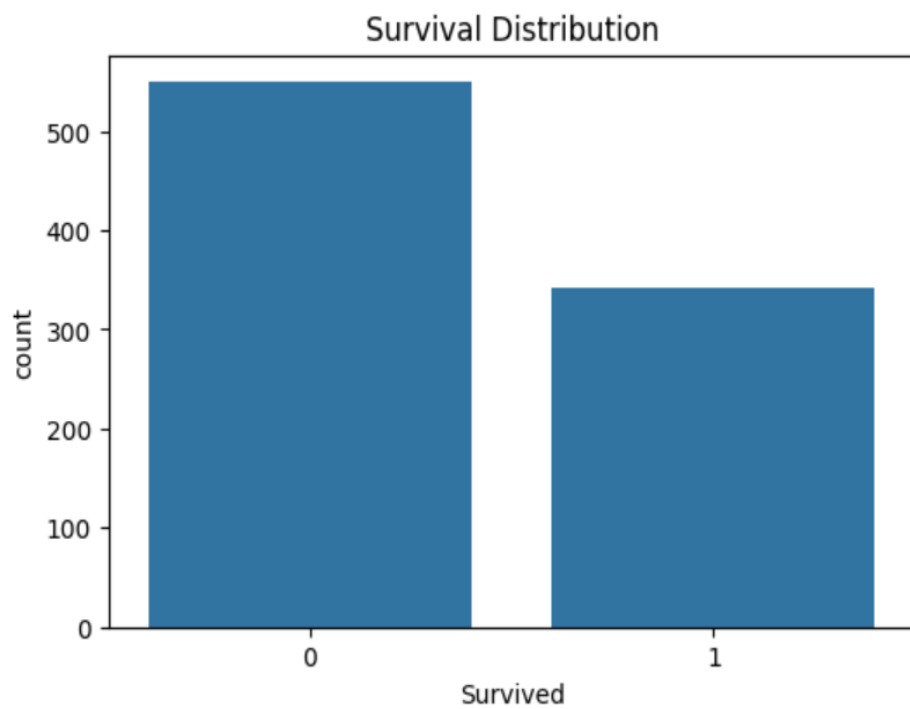
Shows male and female distribution.

Age Distribution



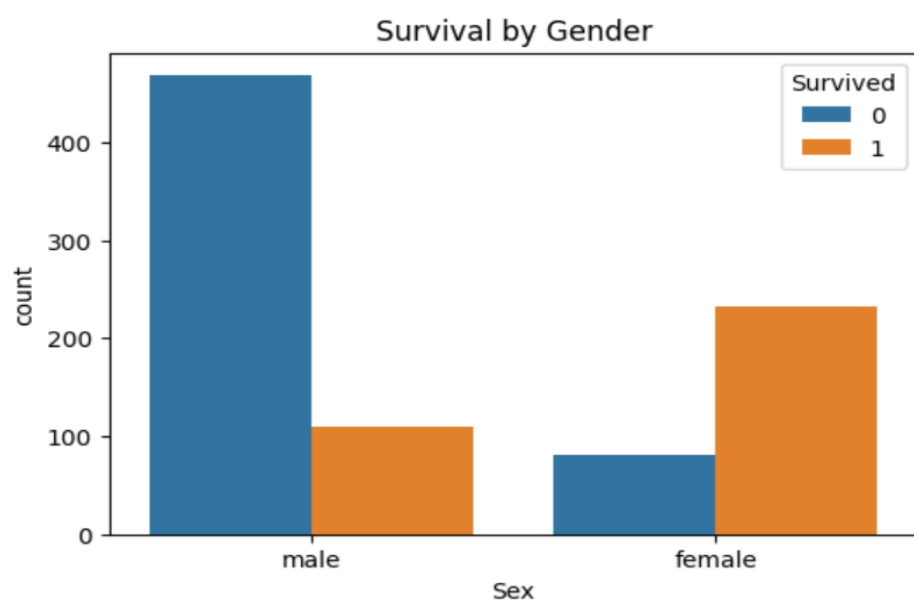
Shows age distribution of passengers.

Survival Distribution



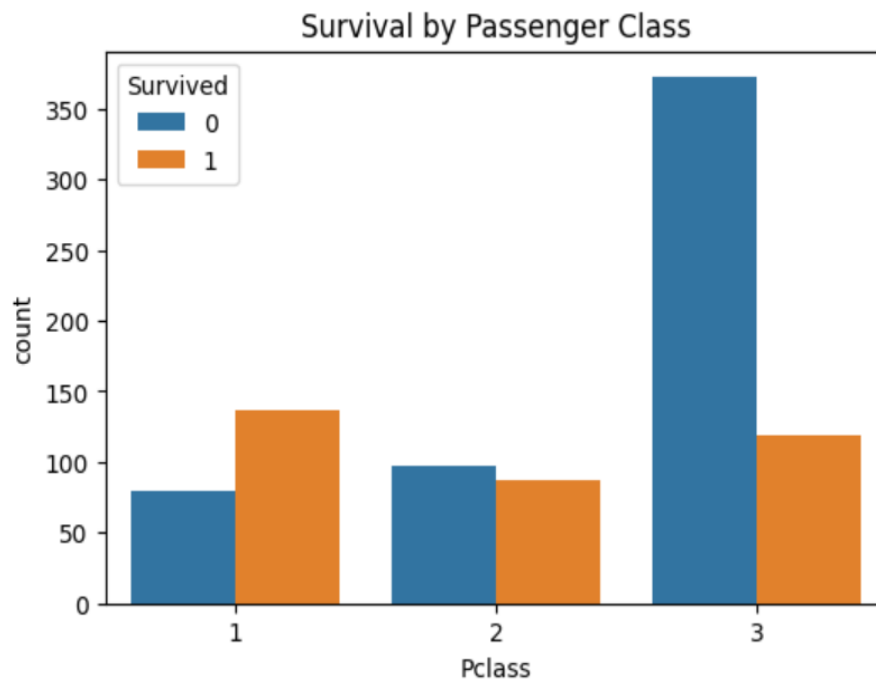
Compares survived vs not survived

Survival by Gender



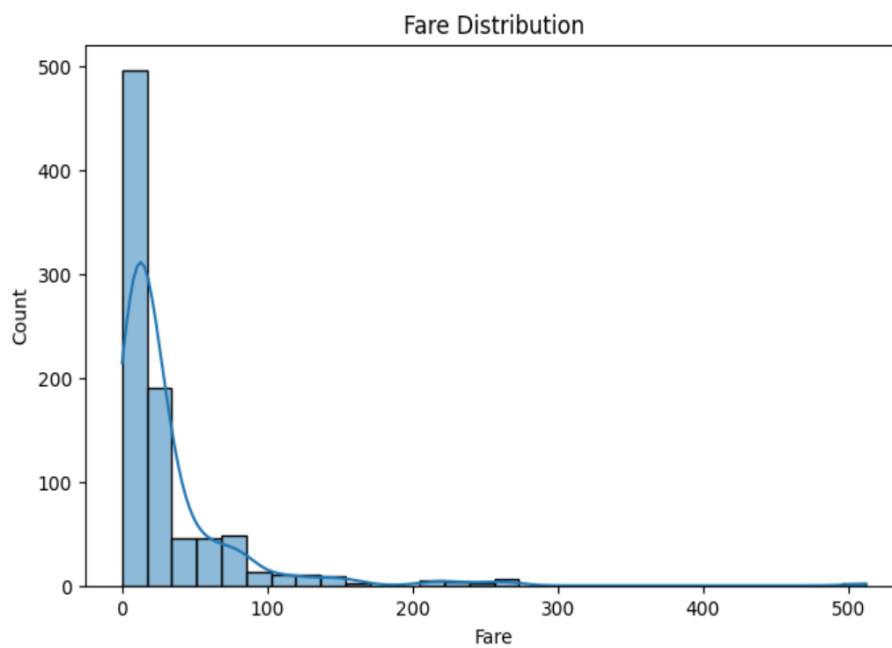
Female passengers had a higher survival rate.

Survival by Passenger Class



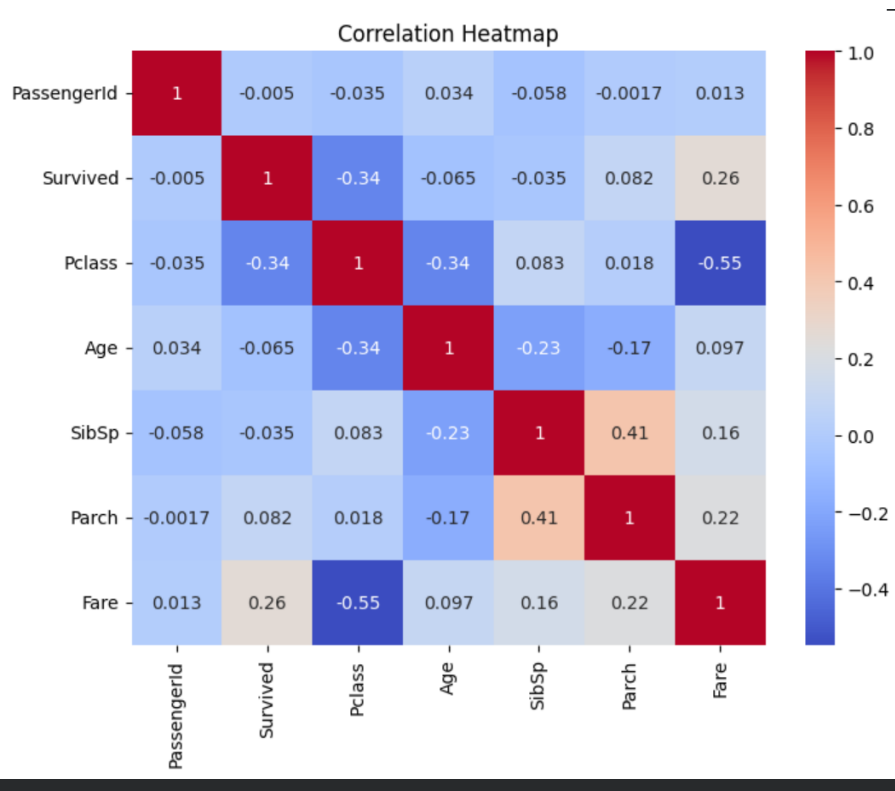
Survival varied by passenger class.

Fare Distribution



Shows ticket fare distribution.

Correlation Heatmap



Displays correlations between numerical features.

Conclusion

The Titanic dataset was successfully explored using statistical summaries and visualizations.